RAMCO INSTITUTE OF TECHNOLOGY
Approved by AICTE, New Delhi & Affiliated to Anna University
NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Regulation 2021

Course Outcome

SEMESTER-1	
HS3152 **Professional English I**	CO1: To use appropriate words in a professional context CO2: To gain understanding of basic grammatic structures and use them in right context CO3: To read and infer the denotative and connotative meanings of technical texts CO4: To write definitions, descriptions, narrations and essays on various topics
MA3151 **Matrices and Calculus**	CO1: Use the matrix algebra methods for solving practical problems. CO2: Apply differential calculus tools in solving various application problems CO3: Able to use differential calculus ideas on several variable functions CO4: Apply different methods of integration in solving practical problems CO5: Apply multiple integral ideas in solving areas, volumes and other practical problems.
PH3151 **Engineering Physics**	CO1: Understand the importance of mechanics CO2: Express knowledge in electromagnetic waves CO3: Demonstrate foundational knowledge in oscillations, optics and lasers CO4: Understand the importance of quantum physics CO5: Apply quantum mechanical principles toward formation of energy bands
CY3151 **Engineering Chemistry**	CO1: Infer the quality of water from parameter data and propose suitable treatment methods CO2: Identify and apply basic nanoscience and nanotechnology concepts in designing nanomaterials CO3: Apply phase rule and composites knowledge for material selection CO4: Recommend suitable fuels for engineering processes CO5: Recognize different energy resources and apply them for appropriate energy sector applications
GE3151 **Problem Solving and Python Programming**	CO1: Develop algorithmic solutions to simple computational problems. CO2: Develop and execute simple Python programs. CO3: Write simple Python programs using conditionals and looping for solving problems.

	CO4: Decompose a Python program into functions. CO5: Represent compound data using Python lists, tuples, dictionaries etc. CO6: Read and write data from/to files in Python programs.
GE3171 **Problem Solving and Python Programming Laboratory**	CO1: Develop diagrammatic solutions to simple computational problems CO2: Execute simple Python programs using Control Structures, functions, strings CO3: Implement various computing strategies for Python-based solutions to real world problems. CO4: Apply compound data using python lists, tuples, sets and dictionaries for basic applications. CO5: Implement read and write data from/to files in python programs. CO6: Utilize Python packages in developing software applications.
BS3171 **Physics and Chemistry Laboratory**	CO1: Understand the functioning of physics laboratory equipment CO2: Use graphical models to analyze laboratory data CO3: Use mathematical models for quantitative reasoning CO4: Access, process and analyze scientific information CO5: Solve problems individually and collaboratively CO6: Analyze quality of water samples (acidity, alkalinity, hardness, DO) CO7: Determine amount of metal ions via volumetric and spectroscopic Techniques. CO8: Analyze and determine the composition of alloys CO9: Learn simple methods of nanoparticle synthesis CO10: Quantitatively analyze impurities using electroanalytical techniques
GE3172 **English Laboratory**	CO1: Listen to and comprehend general and complex academic information CO2: Understand different viewpoints in discussions CO3: Speak fluently and accurately in formal and informal contexts CO4: Describe products and processes clearly and accurately CO5: Express opinions effectively in discussions

SEMESTER-2	
HS3252 **Professional English II**	CO1: Compare and contrast products and ideas in technical texts CO2: Identify and report cause and effect in events and industrial processes through technical texts CO3: Analyse problems, arrive at feasible solutions and communicate them in written format CO4: Present ideas and opinions in a planned and logical manner CO5: Draft effective resumes for job search
MA3251 **Statistics and Numerical Methods**	CO1: Apply testing of hypothesis for small and large samples in real-life problems CO2: Apply concepts of classification and design of experiments in agriculture CO3: Use numerical techniques of interpolation, differentiation and integration for engineering problems

	CO4: Understand and apply methods for solving first and second order ordinary differential equations CO5: Solve partial and ordinary differential equations with initial and boundary conditions for engineering applications
PH3256 **Physics for Information Science**	CO1: Gain knowledge of classical and quantum electron theories and energy band structures CO2: Acquire knowledge of semiconductor physics and its device applications CO3: Understand magnetic properties of materials and their use in data storage CO4: Understand the functioning of optical materials for optoelectronics CO5: Learn basics of quantum structures and their applications
BE3251 **Basic Electrical and Electronics Engineering**	CO1: Use BIS conventions and specifications for engineering drawing CO2: Construct conic curves, involutes and cycloids CO3: Solve problems involving projection of lines CO4: Draw orthographic, isometric and perspective projections of simple solids CO5: Draw the development of simple solids
GE3251 **Engineering Graphics (Environment Studies content given)**	CO1: Summarize the importance of environment, biodiversity, ecosystem and solving environmental problems CO2: Describe causes, effects and control measures of air, water, soil, noise, radioactive and thermal pollution with case studies CO3: Discuss renewable and non-renewable resources and energy conservation CO4: Explain social issues and solutions for sustainable environment with Acts and case studies CO5: Review the impact of human population on the environment and remedial measures
AD3251 **Data Structures Design**	CO1: Explain abstract data types CO2: Design, implement, and analyse linear data structures, such as lists, queues, and stacks, according to the needs of different applications CO3: Design, implement, and analyse efficient tree structures to meet requirements such as searching, indexing, and sorting CO4: Model problems as graph problems and implement efficient graph algorithms to solve them
AD3271 **Data Structures Design Laboratory**	CO1: Implement ADTs as Python classes CO2: Design, implement, and analyse linear data structures, such as lists, queues,and stacks, according to the needs of different applications CO3: Design, implement, and analyse efficient tree structures to meet requirements such as searching, indexing, and sorting CO4: Model problems as graph problems and implement efficient graph algorithms to solve them
GE3271 **Engineering Practices Laboratory**	CO1: Draw pipeline plan; use pipe fittings; perform household plumbing; saw, plane and join wood materials CO2: Wire electrical joints in household electrical systems CO3: Weld joints; perform machining processes (turning, drilling, tapping);

	assemble mechanical components; make sheet metal tray CO4: Solder and test electronic circuits; assemble and test components on PCB
GE3272 Communication Laboratory	CO1: Speak effectively in formal and semi-formal group discussions CO2: Discuss, analyse and present concepts and problems from multiple perspectives CO3: Write emails, letters and job applications CO4: Write critical reports clearly and precisely CO5: Give appropriate instructions and recommendations for safe task execution

SEMESTER-3	
AL3391 Artificial Intelligence	CO1: Explain the intelligent agent frameworks CO2: Apply the problem solving techniques CO3: Apply the game playing and CSP techniques CO4:Discuss the logical reasoning of Intelligent agents using propositional logic CO5:Apply the concept of probabilistic reasoning under uncertainty
AD3301 Data Exploration and Visualization	CO 1: Describe the fundamentals of exploratory data analysis. CO 2: Implement the data visualization using Matplotlib CO 3: Perform univariate data exploration and analysis CO 4: Apply bivariate data exploration and analysis CO 5: Apply Data exploration and visualization techniques for multivariate and time series data. CO 6: Apply Data exploration and visualization techniques for a given data set
AD3391 Database Design and Management	CO1: To learn database development life cycle and conceptual modelling CO2: Apply SQL for data definition, manipulation and design relational databases for applications CO3: Map ER model to Relational model to perform database design effectively and write Queries using normalization criteria. CO4: Appraise the Serializability of any non-serial schedule using concurrency technique CO5: Apply the data model and querying in Object-relational and No-SQL databases. CO6: Construct and implement basic SQL operations and joins in relational databases.
AD3351 Design and Analysis of Algorithms	CO1: Analyze the efficiency of recursive and non-recursive algorithms mathematically CO2: Analyze the efficiency of brute force, divide and conquer, decrease and conquer, Transform and conquer algorithmic techniques CO3: Implement and analyze the problems using dynamic programming and greedy algorithmic techniques.

	CO4: Solve the problems using iterative improvement techniques for optimization. CO5: Compute the limitations of algorithmic power and solve the problems using backtracking and branch and bound techniques
AD3381 **Database Design and Management Laboratory**	CO1: Use typical data definitions and manipulation commands. CO2: Design, implement and test Nested and Join Queries and views. CO3: Analyze the use of Tables, Functions, triggers and Procedures. CO4 Create applications with a Front-end Tool
AD3311 **Artificial Intelligence Laboratory**	CO1: Design, implement basic search strategies – 8-Puzzle, 8 – Queens problem, Cryptarithmetic, A* and Memory Bounded A* algorithms CO2: Implement Minimax algorithm for game playing (Alpha-Beta pruning), Develop algorithms to solve constraint satisfaction problems CO3: Implement propositional model checking algorithms, forward chaining, backward chaining, and resolution strategies CO4: Design, implement Naïve Bayes model, Bayesian networks and perform inferences

SEMESTER-4	
CS3591 **Computer Networks**	CO 1: Explain the basic layers and its functions in computer networks. CO 2: Apply the functions of various transport layer protocols for given network problems. CO 3: Explain the various functions in the network layers and network protocols CO 4: Apply the routing algorithms for a given problem. CO 5: Discuss the working of various Data Link and Physical Layers protocols. CO6: Apply the suitable network layer protocols for a given network problems
AL3452 **Operating Systems**	CO1: Analyze various scheduling algorithms and process synchronization. CO2 : Explain deadlock, prevention and avoidance algorithms. CO3 : Compare and contrast various memory management schemes. CO4 : Explain the functionality of file systems I/O systems, and Virtualization CO5 : Compare iOS and Android Operating Systems
AL 3451 **Machine Learning**	CO1: Explain the basic concepts of machine learning. CO2 : Construct supervised learning models. CO3 : Construct unsupervised learning algorithms. CO4: Evaluate and compare different models
AD3491 **Fundamentals of Data science and Analytics**	CO1:Write python programs to handle data using Numpy and Pandas CO2: Perform descriptive analytics CO3: Perform data exploration using Matplotlib

	CO4: Perform inferential data analytics CO5: Build models of predictive analytics
AD3411 **Data Science & Analytics Laboratory**	CO1: Use typical data definitions and manipulation commands. CO2: Design, implement and test Nested and Join Queries and views. CO3: Analyze the use of Tables, Functions, triggers and Procedures. CO4 Create applications with a Front-end Tool
AD3461 **Machine Learning Laboratory**	CO1: Apply suitable algorithms for selecting the appropriate features for analysis. CO2: Implement supervised machine learning algorithms on standard datasets and evaluate the performance. CO3: Apply unsupervised machine learning algorithms on standard datasets and evaluate the performance. CO4: Build the graph based learning models for standard data sets. CO5: Assess and compare the performance of different ML algorithms and select the suitable one based on the application

SEMESTER-5	
CS3551 **Distributed Computing**	CO1: Explain the foundations of distributed systems CO2: Solve synchronization and state consistency problems CO3: Use resource sharing techniques in distributed systems CO4: Apply working model of consensus and reliability of distributed systems CO5: Explain the fundamentals of cloud computing
CCS335 **Cloud Computing**	CO1: Discuss the design challenges in the cloud. CO2: Apply the concept of virtualization and its types. CO3: Experiment with virtualization of hardware resources and Docker. CO4: Develop and deploy services on the cloud and set up a cloud environment. CO5: Explain security challenges in the cloud environment. CO6: Explain Identity Access Management Architecture and its practices
AD3501 **Deep Learning**	CO1: Explain the basics in deep neural networks CO2: Apply Convolution Neural Network for image processing CO3: Apply Recurrent Neural Network and its variants for text analysis CO4: Apply model evaluation for various applications CO5: Apply auto encoders and generative models for suitable applications
CW3551 **Data and Information Security**	CO1: Explain the basics of data and information security CO2: Discuss about security investigation such as legal, ethical and professional issue and security policies CO3: Apply the various digital signature schemes to simulate different applications CO4: Explain various architecture and operational techniques of Email and IP security

	CO5: Apply the Web security protocols for ECommerce applications CO6: Construct and implement secure web application
AD3002 **Healthcare Analytics**	CO1: Use machine learning and deep learning algorithms for health data analysis CO2: Apply the data management techniques for healthcare data CO3: Evaluate the need of healthcare data analysis in e-healthcare, telemedicine and other critical care applications CO4: Design health data analytics for real time applications CO5: Design emergency care system using health data analysis CO6:Analyse a clinical Data set and predict the regression
AD3511 **Deep Learning Laboratory**	CO1: Apply deep neural network for simple problems CO2: Apply Convolution Neural Network for image processing CO3: Apply Recurrent Neural Network and its variants for text analysis CO4: Apply generative models for data augmentation CO5: Develop real-world solutions using suitable deep neural networks
CCS334 **Big Data Analytics**	CO1: Describe big data and use cases from selected business domains. CO2: Explain NoSQL big data management. CO3: Install, configure and run Hadoop and HDFS. CO4: Perform map-reduce analytics using Hadoop. CO5: Use Hadoop-related tools such as HBase, Cassandra, Pig, and Hive for big data analytics

SEMESTER-6	
CCS345 **Ethics and AI**	CO1: Discuss about morality and ethics in AI CO2: Acquire the knowledge of real time application ethics, issues and its challenges. CO3: Outline importance of ethical harms and ethical initiatives in AI CO4: Explain about AI standards and Regulations like AI Agent, Safe Design of Autonomous and Semi- Autonomous Systems CO5: Explain Concepts of Robot ethics and Morality with professional responsibilities. CO6: Discuss about the societal issues in AI with National and International Strategies in AI
CCS369 **Text and Speech Analysis**	CO1: Explain existing and emerging deep learning architectures for text and speech processing CO2: Apply deep learning techniques for NLP tasks, language modelling and machine translation CO3: Explain coreference and coherence for text processing CO4: Build question-answering systems, chatbots and dialogue systems CO5: Apply deep learning models for building speech recognition and text-to-

	speech systems
CCW331 **Business Analytics**	CO1: Explain the real world business problems and model with analytical solutions. CO2: Identify the business processes for extracting Business Intelligence CO3: Apply predictive analytics for business fore-casting CO4: Apply analytics for supply chain and logistics management CO5: Use analytics for marketing and sales.
CCS368 **Stream Processing**	CO1: Understand the applicability and utility of different streaming algorithms. CO2: Describe and apply current research trends in data-stream processing. CO3: Analyze the suitability of stream mining algorithms for data stream systems. CO4: Program and build stream processing systems, services and applications. CO5: Solve problems in real-world applications that process data streams.
CS3691 **Embedded Systems and IoT**	CO1: Explain the architecture of embedded processors. CO2: Write embedded C programs. CO3: Design simple embedded applications. CO4: Compare the communication models in IOT CO5: Design IoT applications using Arduino/Raspberry Pi /open platform.
OBT351 **Food, Nutrition and Health**	CO1: To be able to understand the nutritional values of the various types of Foods. CO2: To be able to Analyze the role of food in the metabolic activity of the healthy diet CO3: To be able to Infer the BMI calculation and stress related diseases. CO4: To be able to Elaborate the independent decision on the choice of food to prevent life style disorders and diseases CO5: To be able to Assess about the food laws governance CO6:To be able to Compare junk, modified and super foods
CCS333 **Augmented Reality/Virtual Reality**	CO1: Understand the basic concepts of AR and VR CO2: Understand the tools and technologies related to AR/VR CO3: Know the working principle of AR/VR related Sensor devices CO4: Design of various models using modeling techniques CO5: Develop AR/VR applications in different domains
CCS361 **Robotic Process Automation**	CO1: Enunciate the key distinctions between RPA and existing automation techniques and platforms. CO2: Use UIPath to design control flows and work flows for the target process CO3: Implement recording, web scraping and process mining by automation CO4: Use UIPath Studio to detect, and handle exceptions in automation processes CO5: Implement and use Orchestrator for creation, monitoring, scheduling, and controlling of automated bots and processes

SEMESTER-7	
GE3791 **Human Values and Ethics**	CO1: Identify the importance of democratic, secular and scientific values in harmonious functioning of social life CO2: Practice democratic and scientific values in both their personal and professional life. CO3: Find rational solutions to social problems. CO4: Behave in an ethical manner in society CO5: Practice critical thinking and the pursuit of truth.
GE3755 **Knowledge Management**	CO1: Understand the process of acquire knowledge from experts CO2: Understand the learning organization. CO3: Use the knowledge management tools. CO4: Develop knowledge management Applications. CO5: Design and develop enterprise applications
AI3021 **IT in Agricultural System**	CO1: The students shall be able to understand the applications of IT in remote sensing applications such as Drones etc. CO2: The students will be able to get a clear understanding of how a greenhouse can be automated and its advantages. CO3: The students will be able to apply IT principles and concepts for management of field operations. CO4: The students will get an understanding about weather models, their inputs and applications. CO5: The students will get an understanding of how IT can be used for e-governance in agriculture
OMG355 **Multivariate Data Analysis**	CO1: Demonstrate a sophisticated understanding of the concepts and methods; know the exact scopes and possible limitations of each method; and show capability of using multivariate techniques to provide constructive guidance in decision making CO2: Use advanced techniques to conduct thorough and insightful analysis, and interpret the results correctly with detailed and useful information. CO3: Show substantial understanding of the real problems; conduct deep analysis using correct methods; and draw reasonable conclusions with sufficient explanation and elaboration. CO4: Write an insightful and well-organized report for a real-world case study, including thoughtful and convincing details. CO5: Make better business Decisions by using advanced techniques in data analytics.
CCS338 **Computer Vision**	CO1: To understand basic knowledge, theories and methods in image processing and computer vision. CO2: To implement basic and some advanced image processing techniques in OpenCV. CO3: To apply 2D a feature-based based image alignment, segmentation and

	motion estimations. CO4: To apply 3D image reconstruction techniques CO5: To design and develop innovative image processing and computer vision applications
CCS353 **Multimedia Data Compression and Storage**	CO1: Understand the basics of text, Image and Video compression CO2: Understand the various compression algorithms for multimedia content CO3: Explore the applications of various compression techniques CO4: Explore knowledge on multimedia storage on disks CO5: Understand scheduling methods for request streams
OSF351 **Fire Safety Engineering**	CO1: Understand the effect of fire on materials used for construction CO2: Understand the method of test for non-combustibility and fire resistance; and will be able to select different structural elements and their dimensions for a particular fire resistance rating of a building. CO3: To understand the design concept of fire walls, fire screens, local barriers and fire doors and able to select them appropriately to prevent fire spread. CO4: To decide the method of fire protection to RCC, steel, and wooden structural elements and their repair methods if damaged due to fire. CO5: Describe the safety techniques and improve the analytical and intelligence to take the right decision at right time.

SEMESTER-8	
AD3811 **Project Work / Internship**	CO1: Gain Domain knowledge and technical skill set required for solving industry / research problems CO2: Provide solution architecture, module level designs, algorithms CO3: Implement, test and deploy the solution for the target platform CO4: Prepare detailed technical report, demonstrate and present the work